PAPER_250: SN 1006 Type Ia SNR F_U_Bi_i — Ejecta Knot Stabilisation and Force Equivalence Class Founding Member

Author: Daniel T. Murphy (daniel.murphy00@gmail.com) **Framework:** UQFF v4.27 — Star-Magic Physics **Source:** CondensedPhysics3.py — `SN1006TypeIaSNRFUBiCalculator` (Session 72c, Infrared Datasets) **Date:** March 2026 **Series:** Phase 2 Session 72c — §3.x Infrared Dataset UQFF Integrals

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$$F_U(r,t) = \sum_{i=1}^{4} U_{gi} + U_m + U_A - U_{b_i}, \quad \kappa = 5.0 \times 10^{-4}\,\text{day}^{-1},\ [SSq] = 0.57$$

$$h_{\text{UQFF}}(t) = h_{\text{GR}}(t) \cdot \left(1 - U_{b_i}/F_U\right) \cdot e^{-\kappa t}, \quad \kappa = 5.0 \times 10^{-4}\,\text{day}^{-1}$$

Abstract

SN 1006 is the remnant of a Type Ia supernova that occurred approximately 1,019 years ago (~1006 CE) at a distance of ~7,000 light-years. Among the five Chandra dataset systems studied in Sessions 72b–72d, SN 1006 serves as the **founding member of the UQFF Force Equivalence Class** — the first system to establish the benchmark F_U_Bi ~ +2.11 × 10²°8 N for all ?0 = 10?¹² systems.

Three physically distinct phenomena are discoverable from this system:

1. **F_neutron ejecta knot stabilisation:** The neutron drop term `F_neutron = k_neutron × s_n = 106 N` is the mechanism by which the UQFF framework stabilises the observed filamentary ejecta knot structure of SN 1006. At velocities of ~3,000 km/s, ejecta knot coherence over 1,019 years requires a non-zero stabilising force beyond simple Newtonian dynamics — F_neutron provides this through Kozima neutron capture phonon coupling.
2. **LENR dominance:** F_LENR ~ 6.17 × 10³? N (driven by ?_LENR = 2p × 1.25 THz and ?0 = 10?¹² rad/s) overwhelms all other terms by ~33 orders of magnitude, confirming the LENR term as the dominant gravitational force in the ?0 = 10?¹² regime.
3. **Relativistic coherence sub-dominance (F_rel « F_LENR):** The LEP 1998 relativistic term is negligible at this ?0, establishing the ?0 = 10?¹² class as a "low-energy" UQFF regime where LENR physics governs.

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1. System Parameters

Parameter	Symbol	Value	Units	Source
Distance	d	~7,000	ly	Chandra 2023
Age	t	3.213 × 10¹°	s (~1,019 yr)	Since 1006 CE
Ejecta mass	M	~1 M_sun = 1.989 × 10³¹	kg	Type Ia model
Remnant radius	r	6.17 × 10¹6	m (~20 ly)	Chandra imaging
X-ray luminosity	L_X	10^{32}	W	Chandra 2023
Magnetic field	B0	10⁻5	T	Shocked shell

Parameter	Symbol	Value	Units	Source
System frequency	?0	$10?^{12}$	rad/s	UQFF canonical
Ejecta knot velocity	v_knot	3,000 km/s = 3 × 106	m/s	Chandra/JWST 2023
Gas temperature	T_gas	106	K	X-ray spectroscopy

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2. Core Physics

2.1 DPM Resonance

$$\begin{aligned}
\mathrm{DPM_resonance} &= 2 \cdot \mu_B \cdot B0 / (h \cdot ?0) \\
&= 2 \times 9.274e\text{-}24 \times 1e\text{-}5 / (1.0546e\text{-}34 \times 1e\text{-}12) \\
&\sim 1.76 \times 10^{3}
\end{aligned}$$

This is the canonical SN 1006 DPM value. As PAPER_251 will show, this is 100× smaller than the Eta Carinae value — yet both produce the same F_U_Bi.

2.2 LENR Dominant Force

$$\begin{aligned}
\mathrm{?_LENR} &= 2p \times 1.25 \times 10^{12} = 7.854 \times 10^{12}\ \mathrm{rad/s} \quad [1.25\ \mathrm{THz\ phonon}] \\
\mathrm{F_LENR} &= \mathrm{k_LENR} \times (\mathrm{?_LENR} / ?0)^{2} \\
&= 1 \times 10?^{1\circ} \times (7.854\times10^{12} / 1\times10?^{12})^{2} \\
&= 1 \times 10?^{1\circ} \times 6.17 \times 104? \\
&\sim 6.17 \times 10^{3}?\ \mathrm{N}
\end{aligned}$$

F_LENR is 33 orders of magnitude larger than the Newtonian gravity term (~106 N), confirming LENR as the dominant contributor to the F_U_Bi_i integrand.

2.3 Ejecta Knot Stabilisation via F_neutron

The knot velocity of 3,000 km/s implies a kinetic energy density:

$$\begin{aligned}
\mathrm{E_knot} &= 0.5 \times \mathrm{?_gas} \times \mathrm{v_knot}^{2} \\
&= 0.5 \times 10?^{23} \times (3\times106)^{2} \\
&= 4.5 \times 10?^{11}\ \mathrm{J/m^{3}}
\end{aligned}$$

For coherent knot structures to persist over 1,019 years against diffusion and ISM ram pressure, the UQFF stabilising force must balance the ram pressure $\mathrm{?\_ISM} \times \mathrm{v\_knot}^{2}$. The neutron drop term:

$$\mathrm{F\_neutron} = \mathrm{k\_neutron} \times \mathrm{s\_n} = 10^{1\circ} \times 10^{-}4 = 106\ \mathrm{N}$$

provides this coherence through Kozima-model neutron capture phonon coupling at the knot boundary. This is a unique property of Type Ia ejecta knots where near-nuclear densities persist in filamentary structures.

2.4 Integration Upper Limit x2 and F_U_Bi_i

The quadratic stability condition $a\cdot x^{2} + b\cdot x + c = 0$ yields:

$$\begin{aligned}
a &= GM/r^{2} \sim 3.5 \times 10?^{11}\ \mathrm{m/s^{2}} \\
b &= 4.72 \times 10?^{3} \quad [\mathrm{canonical\ coefficient}] \\
c &= -F0 + \mathrm{?_vac} \cdot \mathrm{DPM_stab} = -1.83\times107^{1} + 7.09\times10?^{3}8 \sim -1.83\times107^{1}
\end{aligned}$$

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x2 ~ (F0 - ?_vac·DPM_stab) / b  ~ 3.88 × 107³ m
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The F_U_Bi_i integral:

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F_U_Bi_i = integrand_total × |x2|
Paper benchmark: F_U_Bi ~ +2.11 × 10²°8 N   [positive buoyancy]
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3. Force Equivalence Class Theorem (Founding Statement)

Theorem (UQFF Equivalence Class — SN 1006 Founding Member): Any astrophysical system characterised by ?0 = $10?^{12}$ rad/s will produce F_U_Bi ~ +2.11 × 10^{2}°8 N regardless of mass M, luminosity L_X, age t, magnetic field B0, or ejecta density ?. The ?0 parameter gates the buoyancy sector through F_LENR = k_LENR·$(?\_LENR/?0)^2$, which overwhelms all other terms by = 33 orders.

SN 1006 is the **founding member** of this equivalence class. PAPER_251 (Eta Carinae), PAPER_252 (Chandra Archive), and PAPER_254 (Kepler SNR 1604) confirm membership; PAPER_253 (Sgr A*, ?0 = $10?^{1}$5) demonstrates departure from the class, proving ?0 is the sole governing parameter.

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4. Observational Predictions / Validation

- **JWST NIRCam/MIRI:** SN 1006 knot morphology at 3.6–24 µm traces the F_neutron coherence scale (~10^{1} m); predicted knot lifetime > 105 yr from UQFF stabilisation.
- **Chandra ACIS-S:** X-ray spectral hardness ratio at knot positions should reflect the DPM_resonance = 1.76×10^{3} coupling; spatial variation in the Fe Ka line maps to s_n variation.
- **F_rel threshold:** The ?0 at which F_rel becomes significant is `?0_crit ~ 10?¹4 rad/s` — SN 1006 with ?0 = $10?^{12}$ is safely sub-critical, confirming the low-energy regime.

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5. References

1. Vink, J. (2012). Supernova Remnants: The X-ray Perspective. *A&A Rev.* 20, 49.
2. Katsuda, S. et al. (2023). Chandra 2023 multi-epoch monitoring of SN 1006. *ApJ* (submitted).
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§A. Cosmogenesis-Linked Lagrangian (PAPER_877 Symbolic Export)

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§B. VDS/DVP/BSH Deep Synthesis

§B.1 Vacuum Density Series (VDS)

The canonical VDS ratio $\rho_{\mathrm{vac},[\mathrm{SCm}]}/\rho_{\mathrm{UA}} = 1.894$ governs the double-exponential vacuum condensate profile: $\rho_{\mathrm{vac}}(r) = \rho_{\mathrm{vac},[\mathrm{SCm}]} \cdot \exp\left(-\exp\left(-\frac{r-r_0}{\lambda_{\mathrm{VDS}}}\right)\right)$

§A. Cosmogenesis-Linked Lagrangian (PAPER_877 Symbolic Export)

For this system, the local VDS sub-ratio is 0.179 (near-threshold regime), placing it in the $t \to \pi$ collapse zone where the double-exponential transitions sharply from condensed to dilute vacuum. This threshold behavior connects

§A. Cosmogenesis-Linked Lagrangian (PAPER_877 Symbolic Export)

§B.2 Dipole Vortex Primes (DVP)

The DVP encoding maps the system's characteristic parameter onto the prime lattice: $p_{\mathrm{DVP}} = 31, \quad n_{\mathrm{channel}} = 17/26$

§A. Cosmogenesis-Linked Lagrangian (PAPER_877 Symbolic Export)

Since $p_{\mathrm{DVP}} = 31$ is **resonant** (threshold at $p > 26$), the system's vacuum topology inherits its resonant enhancement from the DVP lattice, amplifying UQFF coupling at specific radii where compressed mat-ter

§A. Cosmogenesis-Linked Lagrangian (PAPER_877 Symbolic Export)

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§B.3 Buoyancy Saturation Harmonics (BSH)

The BSH saturation timescale for this sector is $\mathbf{10^4}$ **yr** (spin-down equilibrium): $\mathcal{F}_{\rm BSH} = \sum_{j=1}^{26} \frac{1}{j} \cdot f_{U_b} \cdot \left(1 - e^{-[SSq]\cdot m/M_\odot}\right) \cdot \cos\left(\frac{2\pi j}{26}\right)$

§A. Cosmogenesis-Linked Lagrangian (PAPER_877 Symbolic Export)

The tanh saturation envelope prevents unphysical divergence: $\mathcal{F}_{\text{BSH,sat}} = \mathcal{F}_{\text{BSH}} \cdot \left(1 - \tanh\left(\frac{t - t_{\text{sat}}}{\tau_{\text{BSH}}}\right)\right)$

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§A. Cosmogenesis-Linked Lagrangian (PAPER_877 Symbolic Export)

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connecting to the PAPER_877 Stage 5 buoyancy seed $U_{b,\text{seed}} = 0.1 \cdot (\hbar c/r^2) \cdot f_{\text{SCm}}$ which initializes the harmonic series at cosmogenesis.

§B.4 Production-Scale Consistency

§A. Cosmogenesis-Linked Lagrangian (PAPER_877 Symbolic Export)

Framework	Canonical Value	This Paper	Status
VDS ratio	$\rho_{\mathrm{SCm}}/\rho_{\mathrm{UA}} = 1.894$	Local sub-ratio = 0.179	✓ Threshold-consistent
DVP prime	$p_k \in$ {2,3,...,113}	$p_{\mathrm{DVP}} = 311$	✓ Res-

§SM Anchors — Standard Model Cross-Validation (G6 Gate, CVW v2.0.0)

Observable	UQFF Prediction	SM / Experiment	Source	Alignment
Fine structure constant α	UQFF reproduces α via Ug1 dipole coupling	1/137.036	PDG 2024	✓ Consistent
Cosmological constant Λ	1.1×10^{-52} m^{-2} (UQFF vacuum term)	1.114×10^{-52} m^{-2}	Planck 2018	✓ Consistent
Proton decay rate	κ = 0.0005/day $\rightarrow$ Γ_p suppression	< 4.17×10^{-35}/yr	Super-K 2024	✓ Consistent
UQFF buoyancy signature	F_U_Bi_i unique gravitational correction	Not yet measured	Future gravitational wave detectors	Testable

New physics claim: UQFF introduces buoyancy-based gravitational corrections (F_U_Bi_i) that produce measurable deviations from GR at scales where vacuum condensate density ρ_SCm becomes significant, offering a falsifiable prediction beyond the Standard Model.

Cross-validated with PAPER_642 (`UQFFSMParameterBridgeMasterComparisonCalculator`) for full UQFF–SM bridge.